

Reliable LLM Reasoning under False Premises and Unreliable Verification

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Research Focus

My research focuses on building reliable reasoning systems that remain robust when their inputs contain false premises, hallucinated assumptions, or unreliable verification signals. My central view is that LLM failures are not only outcome-level errors, but often arise from structured failures across the reasoning process. I study how unreliable semantics enter the context, route the generator toward fragile behaviors, and contaminate the selector that is supposed to verify or correct the final answer.

Unifying Framework: Context, Generator, and Selector

I organize reliable LLM reasoning into three layers. The **Context** layer asks whether the model should trust semantic information from prompts, retrieved documents, tool outputs, or previous generations. The **Generator** layer asks how unreliable semantics change reasoning behavior, such as whether the model accepts, repairs, or ignores a false premise. The **Selector** layer asks whether test-time evidence, verifier feedback, or candidate selection signals are independent from the original failure mode. A reliable reasoning system should not only produce correct final answers, but also model context reliability, maintain behavior robustness, and preserve evidence independence.

Recent Manuscripts

Understanding Test-Time Selection Failures under False Premises in Code Generation *ACL ARR 2026 May Submission*.

[Google Drive PDF](#)

This work studies the **Selector** layer. Across 3 code-generation benchmarks and 5 code models, we show that false premises reduce selected correctness even after 64-sample test-time selection. The largest selected-correctness drop reaches **13.81 pp**; verifier-written tests inherit the same premise-level blind spot, reducing counterexample coverage by 18.9 pp. A lightweight gold-free predictor reaches 0.846 AUROC, and decoupled auditor scaling recovers +5.40 pp.

Locked Behavior: Understanding False-Premise Failures in Coding *NeurIPS 2026 Submission*.

[Google Drive PDF](#)

This work studies the **Generator** layer. Across HumanEval and MBPP with 4 representative code models, false premises consistently reduce pass rates, with false sortedness causing the largest drop of **23.17 pp**. Using Controlled Behavior Evaluation, we track three solution routes, ACCEPT, REPAIR, and IGNORE; a counterfactual decomposition shows that resetting only ACCEPT recovers 64.88% of the observed drop.

LLM Hallucinations / Post-Hallucination Reasoning *ECCV 2026 Submission*.

[Google Drive PDF](#)

This work studies the **Context** layer. Across 9 tasks and 9 models, HIVE reveals structured modality-dependent patterns: hallucinated captions often improve accuracy on vision-language tasks, while text-only tasks show limited or unstable effects. The analysis suggests that hallucinated semantics can act as controlled semantic interventions, broadening semantic coverage and reshaping reasoning dynamics while preserving stable inference.

Future Direction

My next step is to move from diagnosing these failures to constructing reliable reasoning systems. At the **Context** layer, I want to develop semantic reliability modeling so that LLMs can identify which inputs, retrieved evidence, tool outputs, or intermediate agent states deserve trust. At the **Generator** layer, I want to study behavior-robust generation, where models learn not to over-accept misleading premises and instead inspect, repair, or reject unreliable assumptions. At the **Selector** layer, I want to build evidence-decoupled test-time reasoning systems, where verification, self-correction, and agent feedback are evaluated for independence from the original failure mode. More broadly, I hope to develop LLM agents that reason reliably under flawed assumptions, ambiguous environments, and unreliable feedback.

Research Fit

This agenda naturally connects to summer research on LLM agents, code generation, tool use, verification, and reliable reasoning. I am especially interested in studying how agentic systems should allocate trust across prompts, retrieved evidence, tool outputs, self-generated critiques, and verifier feedback when the reasoning environment itself may contain flawed assumptions.